CSC61604 Computer Networks

Cover Page for Group Assignment

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Table of Contributions

Name	Tasks	Contribution Percentage
Priyalakshimi A/P Saravanan	- Device Placement - Network Topology - Configured details - Troubleshooting	25%
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1.0 Introduction

This report outlines the network design and configuration of a Smart Co-Working Space for a two-storey office facility that consists of a ground floor and a first floor. The network is designed to accommodate a variety of user groups, such as employees, private office tenants, freelancers who will be using the hot desk area, and guests. Other than that, the facility also uses Internet of Things (IoT) technologies that need a dependable and organized network infrastructure, such as wireless projectors, smart door locks, security cameras, and smart lights.

The objective of the suggested design is to maintain reliable network security and performance while offering both wired and wireless connectivity. VLAN segmentation is used to divide several user groups, such as IoT devices, tenants, administrations, and guest WiFi users. This means it guarantees the protection of critical and sensitive internal resources while it is still enabling guests and tenants to access the internet and the required services.

The application that is used for this assignment is Cisco Packet Tracer to demonstrate the proposed network implementation. This report includes the physical floor layout, network topology diagram, IP addressing scheme, VLAN configuration, and connectivity verification. This is to ensure the network supports reliable communication between devices and external server connections.

2.0 Physical Space Layout

2.1 Ground Floor

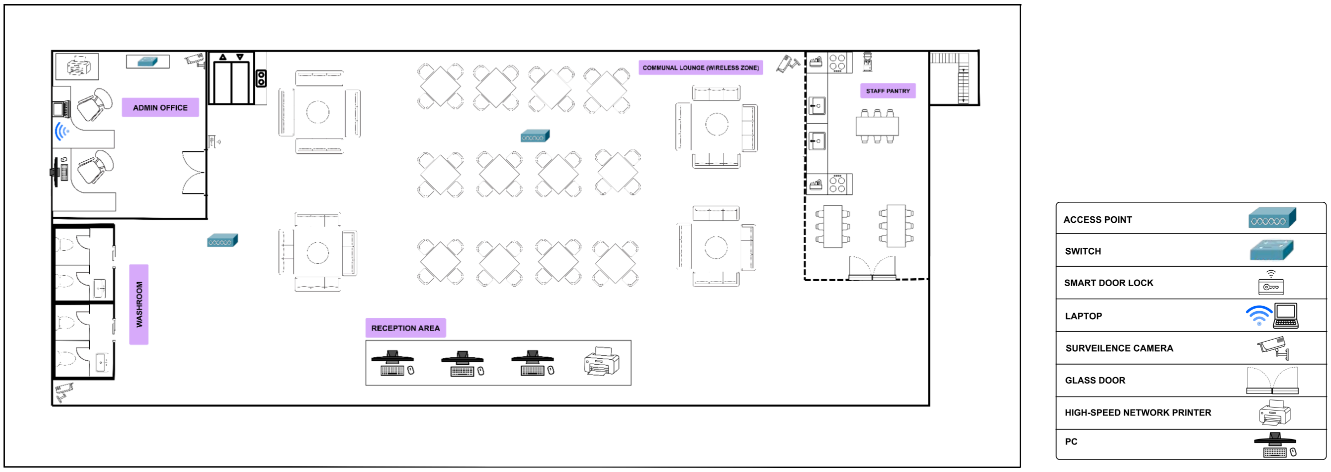

Figure 2.1 Ground Floor Layout

The figure above shows the Ground Floor layout, which consists of three main areas: the Reception Area, Admin Office, and Communal Lounge for guests. The network design supports both wired and wireless connectivity to ensure flexibility. There are 3 wired PCs for receptionists and 1 network printer located in the reception area, which is a wired zone. The Admin Office contains 1 wired PC, 1 laptop connected through Wi-Fi, and 1 network printer dedicated to admin use. The Communal Lounge is a wireless zone where guests and visitors can connect to the network through Wi-Fi access points, with restricted access for up to 15 users on the ground floor.

2.2 First Floor

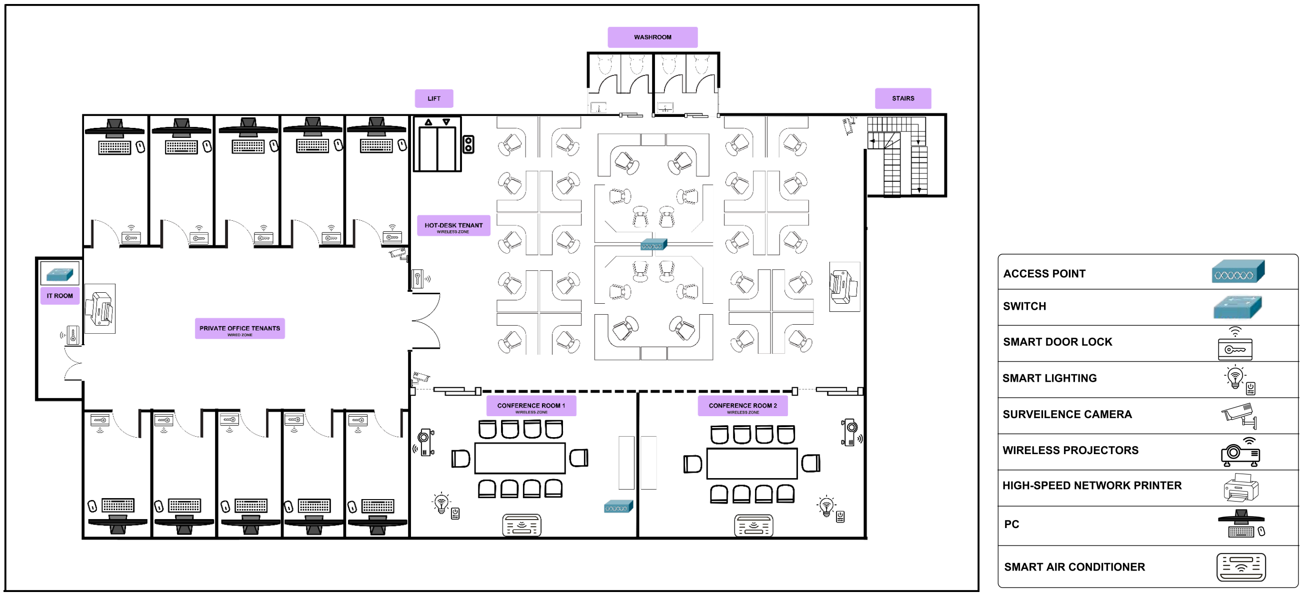

Figure 2.1 First Floor Layout

The figure above illustrates the first floor layout plan, which consists of four main areas: the Private Office Tenants area, the Hot Desk Tenant area, and Conference Rooms 1 and 2. The floor is designed to support both wired and wireless networking. The Private Office Tenants area is a wired zone that includes one IP surveillance camera, ten wired PCs, one dedicated high-speed network printer, and one smart door lock. An IT room with a smart door lock is also located within this area. The Hot Desk Tenant area is designed as a wireless zone and includes one access point, one dedicated high-speed network printer, two IP surveillance cameras, and multiple seating spaces. Lastly, Conference Rooms 1 and 2 are wireless zones equipped with smart lighting, wireless projectors, smart air conditioners, meeting tables, and projector screens. A network switch is placed in Conference Room 1.

3.0 Network Topology

The Network Topology is designed by using a hierarchical topology structure that is based on a three-layer hierarchy. This hierarchy includes the Core layer, the Distribution layer, and the Access layer. This hierarchical structure increases the scalability, performance, and reliability of the network. Besides that, the network is segmented into four different VLANs:

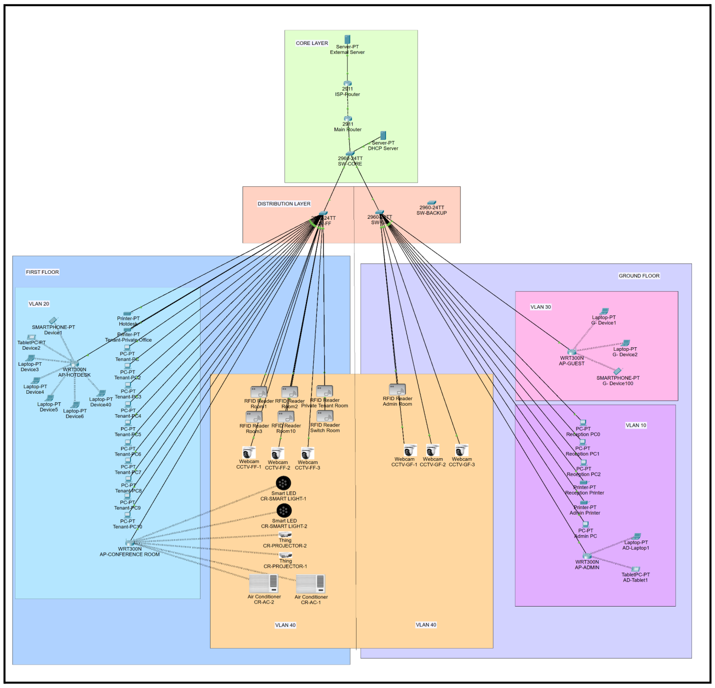

Figure 3.0 Overall Network Topology

- VLAN 10: Administration (Staff PCs & Printers).
- VLAN 20: Premium Tenants and Hot-Desk members (Private Office PCs, freelancers & Tenant Printers).
- VLAN 30: Guest Wi-Fi.
- VLAN 40: IoT Management (Cameras, Door Locks, Projectors).

Having multiple VLANs ensures the separation of different types of network traffic, increasing network security by separating network users, reducing network congestion by limiting broadcasts in each VLAN, and making it easier to manage the network by controlling the different types of network traffic.

3.1 Core Layer

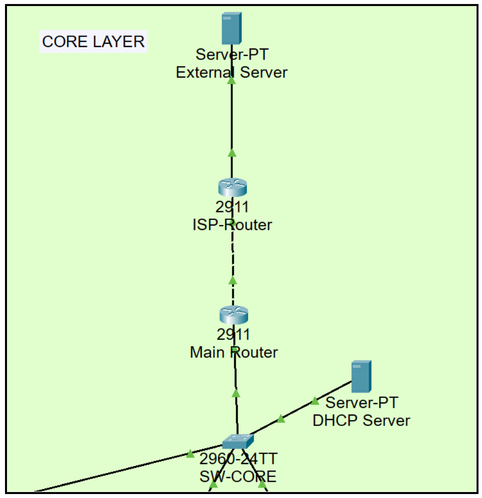

Figure 3.1 Core Layer

The core layer connects the internal network to external services. The Main Router links to the ISP Router which provides connectivity to the external server. The Main Router also connects to the switch core (SW-CORE), which distributes network traffic to the rest of the network. A DHCP server is connected to the switch core to automatically assign IP addresses to devices.

3.2 Distribution Layer

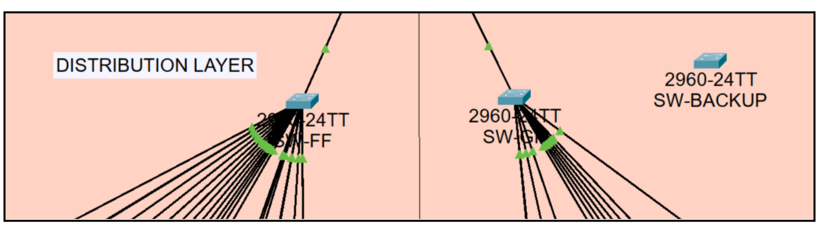

Figure 3.2 Distribution Layer

The distribution layer connects access switches to the core network. SW-FF represents the first-floor switch, while SW-GF represents the ground-floor switch. Both switches aggregate connections from end devices and its connected to the core switch (SW-CORE). A backup switch (SW-BACKUP) is included to provide redundancy in case the core switch fails.

3.2 Access Layer

3.2.1 Ground Floor

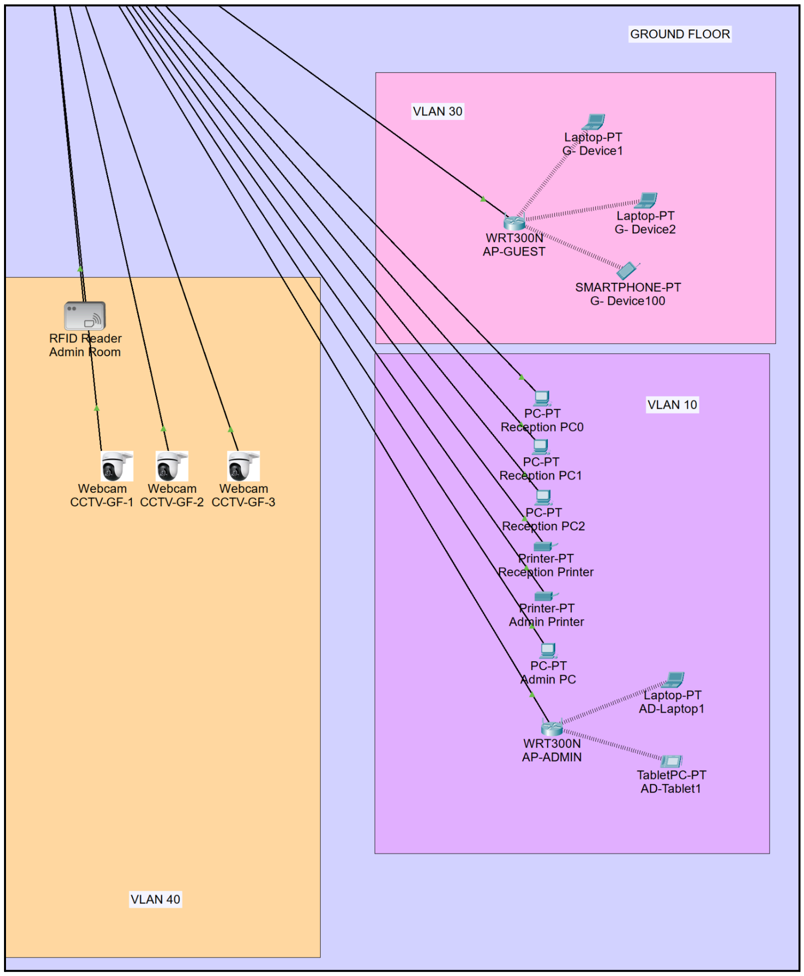

Figure 3.2.1 Topology of Ground Floor

The Ground Floor Access Layer provides connectivity for the end devices and IoT devices, through the access switch. The devices on the floor include three reception PCs, one admin PC, one admin laptop, and one reception printer, and one admin printer. The reception PCs, admin PC, and printers are connected using the wired connections, while the admin laptop is connected using the wireless connection through the AP-ADMIN access point. For the guests, there's a separate wireless access point, AP-GUEST. There are three CCTVs and one RFID reader located in the ground floor area, which are connected using the access switch.

3.2.2 First Floor

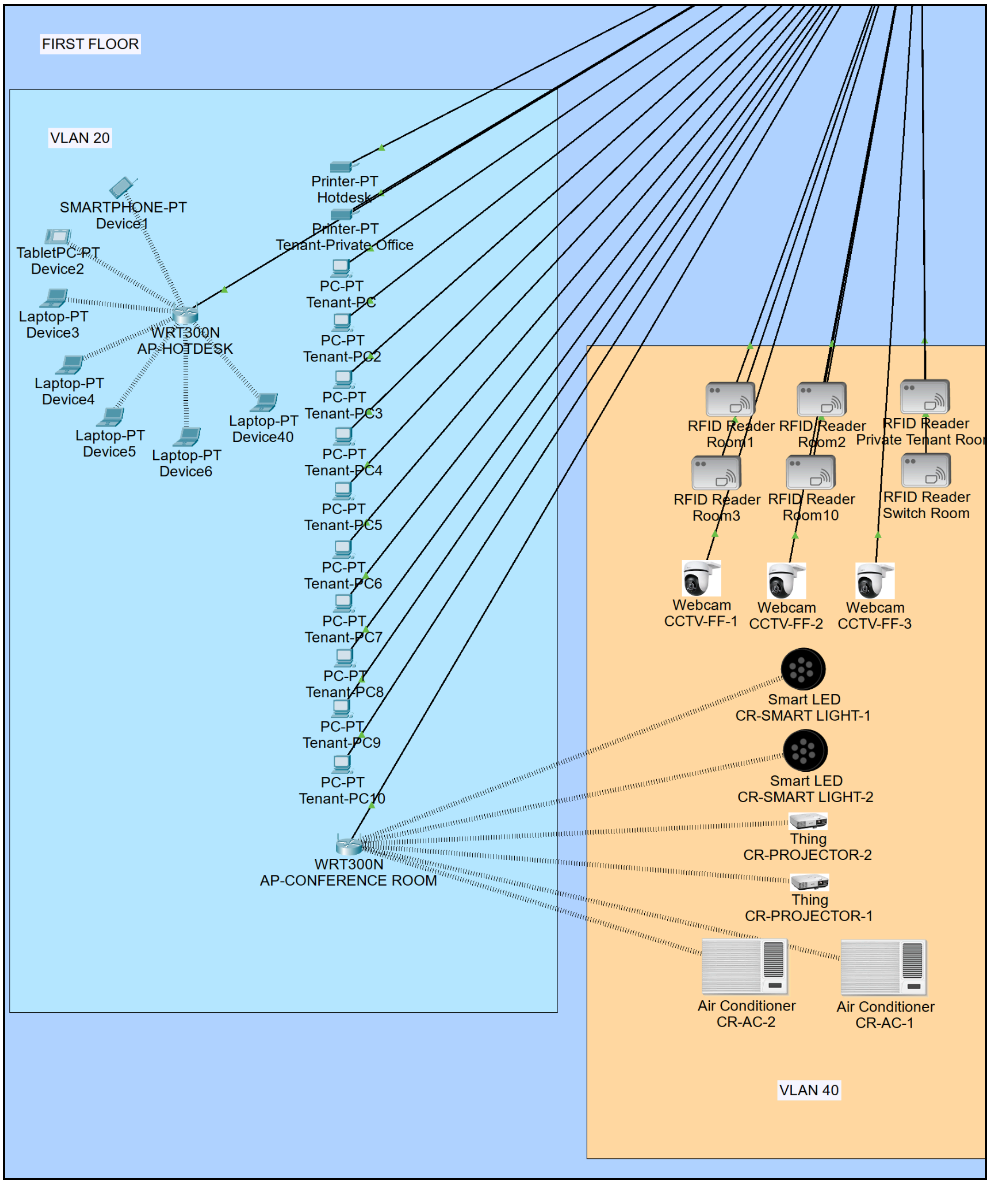

Figure 3.2.2 Topology of First Floor

The first-floor access layer connects wired devices (ten PCs, two printers) in the private office area, wireless users through two APs (HOT DESK, CONFERENCE ROOM), and various IoT/service devices. IoT devices include three CCTVs and two RFID readers (switch room, private room), two smart LED lights, and two smart projectors in the conference rooms.

4.0 IP Addressing Scheme

The 192.168.x.x/24 IP addressing scheme was selected because it follows the RFC 1918 private address range standard. These addresses are not routable on the public internet, which provides built-in security (Anghel et al., 2025).

4.1 VLAN and Subnet Allocation Plan
Each VLAN has its own subnet and default gateway as follows :

VLAN	VLAN Name	Subnet	Default Gateway	IP Assignment Type
10	Administration	192.168.10.0/24	192.168.10.1	DHCP
20	Premium Tenants & Hot-Desk	192.168.20.0/24	192.168.20.1	DHCP (clients), Static (printers/APs)
30	Guest	192.168.30.0/24	192.168.30.1	DHCP
40	IoT Management	192.168.30.0/24	192.168.40.1	Static

Table 4.1: VLAN and Subnet Allocation Plan

4.2 VLAN Deployment and Port Assignments by Switch

Switch Name	Floor	VLAN 10	VLAN 20	VLAN 30	VLAN 40
SW-CORE	Core/Distribution	N/A	N/A	N/A	Fa0/3
SW-GF	Ground	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7	N/A	Fa0/8, Fa0/9	Fa0/10, Fa0/11, Fa0/12, Fa0/13
SW-FF	First	N/A	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/20	N/A	Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19
SW-BACKUP	Backup	N/A	N/A	N/A	N/A

Table 4.2: VLAN Deployment and Port Assignments by Switch

4.3 WAN and External Server Addressing

Device/Interface	IP Address	Subnet Mask	Purpose	Default Gateway
ISP-Router G0/1	10.0.0.1	255.255.255.252	WAN link to Main Router	N/A
Main Router G0/1	10.0.0.2	255.255.255.2	WAN link to ISP-Router	10.0.0.1
ISP-Router G0/2	203.0.113.1	255.255.255.0	Connection to External Server	N/A
External Server	203.0.113.88	255.255.255.0	External Server (Internet)	203.0.113.1
ISP-Router G0/0	203.0.113.1	255.255.255.0	Not in use (Administratively down)	N/A

Table 4.3: WAN and External Server Addressing

4.4 Router Subinterface Addressing

Subinterface	VLAN	Encapsulation	IP Address	Purpose
GigabitEthernet0/0.10	10	dot1Q	192.168.10.1/24	Administration
GigabitEthernet0/0.20	20	dot1Q	192.168.20.1/24	Tenants & Hot-Desk
GigabitEthernet0/0.30	30	dot1Q	192.168.30.1/24	Guest WiFi (ACL applied)
GigabitEthernet0/0.40	40	dot1Q	192.168.40.1/24	IoT Management

Table 4.4: Router Subinterface Addressing

4.5 Device IP Address

We assign each device an IP address based on a plan. The plan takes into account the need for static IP addresses in infrastructure and the need for dynamic IP addresses for end users. For each VLAN subnet, the first ten IP addresses (192.168.x.2 to 192.168.x.10) are reserved as static addresses for network devices such as access points. End-user devices, such as PCs, laptops, and smartphones, obtain their IP addresses using DHCP from the main range (192.168.x.11 to 192.168.x.254) in a dynamic addressing.

The following subsections detail the IP address assignments for each VLAN. They specify which devices use static addresses and which use dynamic addressing.

4.5.1 Administration VLAN

- Subnet: 192.168.10.0/24
- Default Gateway: 192.168.10.1

Device	IP Address	IP Assignment
Reception PC (PC0 - 2)	192.168.10.11 – 192.168.10.254 (excluding 192.168.10.50 & 192.168.10.51)	DHCP
Reception Printer	192.168.10.50	Static
Admin Printer	192.168.10.51	Static
Admin PC, Laptop, Tablet	192.168.10.11 – 192.168.10.254 (excluding 192.168.10.50 & 192.168.10.51)	DHCP
AP-Admin (Access Point)	192.168.10.2	Static

Table 4.5.1: IP Address Allocation for Administration VLAN (VLAN 10)

4.5.2 Premium Tenant & Hot-Desk Members VLAN

- Subnet: 192.168.20.0/24
- Default Gateway: 192.168.20.1

Device	IP Address	IP Assignment
Hotdesk Devices (1-40)	192.168.20.11 to 192.168.20.254 (excluding 192.168.20.50 & 192.168.20.51)	DHCP
Hotdesk Printer	192.168.20.51	Static
Tenant Private Office Printer	192.168.20.50	Static
Tenant-PC (1-10)	192.168.20.11 to 192.168.20.254 (excluding 192.168.20.50 & 192.168.20.51)	DHCP
AP-HOTDESK (Access Point)	192.168.20.2	Static

Table 4.5.2: IP Address Allocation for Premium Tenant & Hot-Desk VLAN (VLAN 20)

4.5.3 Guest VLAN

- Subnet: 192.168.30.0/24
- Default Gateway: 192.168.30.1

Device	IP Address	IP Assignment
AP-GUEST (Access Point)	192.168.30.2	Static
Guest Wireless Devices (1-100)	192.168.30.11 – 192.168.30.254	DHCP

Table 4.5.3: IP Address Allocation for Guest VLAN (VLAN 30)

4.5.4 IOT VLAN

- Subnet: 192.168.40.0/24
- Default Gateway: 192.168.40.1

Device	IP Address	IP Assignment
Private Tenant Room RFID Reader (Main Door)	192.168.40.22	Static
Private Tenant Room1 RFID Reader	192.168.40.50	
Private Tenant Room2 RFID Reader	192.168.40.51	
Private Tenant Room3 RFID Reader	192.168.40.53	
Private Tenant Room10 RFID Reader	192.168.40.54	
Switch Room RFID Reader	192.168.40.21	
Admin Room RFID Reader	192.168.40.21	
Smart Light 1	192.168.40.40	
Smart Light 2	192.168.40.41	
Projector 1	192.168.40.100	
Projector 2	192.168.40.101	
Smart AC 1	192.168.40.30	
Smart AC 2	192.168.40.31	
Ground Floor CCTV 1	192.168.40.13	
Ground Floor CCTV 2	192.168.40.14	
Ground Floor CCTV 3	192.168.40.15	
First Floor CCTV 1	192.168.40.16	
First Floor CCTV 2	192.168.40.11	
First Floor CCTV 3	192.168.40.12	

Table 4.5.2: IP Address Allocation for IoT Management Devices VLAN (VLAN 40)

5.0 Configuration Details

5.1 VLANs Configuration

5.1.1 VLAN Creation

VLANs 10, 20, 30, and 40 were all created on all three switches (SW-CORE, SW-GF, SW-FF) by using the following commands:

```
SW-CORE>enable
SW-CORE#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#vlan 10
SW-CORE(config-vlan)#name Administration
SW-CORE(config-vlan)#exit
SW-CORE(config)#vlan 20
SW-CORE(config-vlan)#name Tenants
SW-CORE(config-vlan)#exit
SW-CORE(config)#vlan 30
SW-CORE(config-vlan)#name Guest-WiFi
SW-CORE(config-vlan)#exit
SW-CORE(config)#vlan 40
SW-CORE(config-vlan)#name IoT_Management
SW-CORE(config-vlan)#exit
SW-CORE(config)#end
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5.1.1.1 VLAN Creation in SW-CORE CLI

5.1.2 Access Port Assignments

Access Ports were configured on SW-GF and SW-FF to assign end divides to their respective VLANs using the following commands:

SW-GF (Ground Floor):

```
SW-GF>enable
SW-GF#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-GF(config)#interface range fastEthernet 0/2-8
SW-GF(config-if-range)#switchport mode access
SW-GF(config-if-range)#switchport access vlan 10
SW-GF(config-if-range)#exit
SW-GF(config)#interface fastEthernet 0/9
SW-GF(config-if)#switchport mode access
SW-GF(config-if)#switchport access vlan 30
SW-GF(config-if)#exit
SW-GF(config)#interface range fastEthernet 0/10-13
SW-GF(config-if-range)#switchport mode access
SW-GF(config-if-range)#switchport access vlan 40
SW-GF(config-if-range)#exit
```

Figure 5.1.2.1: Access Port Configuration in SW-GF CLI

SW-FF (First Floor):

```
SW-FF>enable
SW-FF#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-FF(config)#interface range fastEthernet 0/2-14
SW-FF(config-if-range)#switchport mode access
SW-FF(config-if-range)#switchport access vlan 20
SW-FF(config-if-range)#exit
SW-FF(config)#interface range fastEthernet 0/16-20
SW-FF(config-if-range)#switchport mode access
SW-FF(config-if-range)#switchport access vlan 40
SW-FF(config-if-range)#exit
SW-FF(config)#exit
```

Figure 5.1.2.2: Access Port Configuration in SW-FF CLI

5.1.3 Trunk Port Configuration

Trunk ports were configured on the SW-CORE to carry multiple VLANs between switches and the Main Router using the following commands:

SW-CORE Trunk Configuration:

```
SW-CORE>enable
SW-CORE#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#interface fastEthernet 0/1
SW-CORE(config-if)#switchport mode trunk
SW-CORE(config-if)#switchport trunk allowed vlan 10,20,30,40
SW-CORE(config-if)#no shutdown
SW-CORE(config-if)#exit
SW-CORE(config)#interface fastEthernet 0/2
SW-CORE(config-if)#switchport mode trunk
SW-CORE(config-if)#switchport trunk allowed vlan 10,20,30,40
SW-CORE(config-if)#no shutdown
SW-CORE(config-if)#exit
SW-CORE(config)#interface gigabitEthernet 0/1
SW-CORE(config-if)#switchport mode trunk
SW-CORE(config-if)#switchport trunk allowed vlan 10,20,30,40
SW-CORE(config-if)#no shutdown
SW-CORE(config-if)#exit
```

Figure 5.1.3.1: Trunk Configuration in SW-CORE CLI for SW-GF, SW-FF and Main Router

SW-GF Trunk Configuration:

```
SW-GF>enable
SW-GF#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-GF(config)#interface fastEthernet 0/1
SW-GF(config-if)#switchport mode trunk
SW-GF(config-if)#switchport trunk allowed vlan 10,20,30,40
SW-GF(config-if)#no shutdown
SW-GF(config-if)#exit
```

Figure 5.1.3.2: Trunk Configuration in SW-GF for SW-CORE

SW-FF Trunk Configuration:

```
SW-FF>enable
SW-FF#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-FF(config)#interface fastEthernet 0/1
SW-FF(config-if)#switchport mode trunk
SW-FF(config-if)#switchport trunk allowed vlan 10,20,30,40
SW-FF(config-if)#no shutdown
SW-FF(config-if)#exit
```

Figure 5.1.3.2: Trunk Configuration in SW-FF CLI for SW-CORE

5.2 Static IP Assignment

The static IP addresses were assigned to infrastructure devices to ensure a consistent addressing. This means it will allow devices such as printers, access points, and IoT devices to always be reachable at the same IP address by other network devices. The configuration was done through the Graphical User Interface (GUI) for each device.

5.2.1 Printers

To configure the network printers with static IP addresses in Packet Tracer, the process involves selecting the printer and going to the Config tab. Then, the FastEthernet0 interface is selected. The IP Configuration should be set to Static. The IPv4 address should be set according to the VLAN to which the printer belongs (e.g., 192.168.x.50 or 1), the Subnet Mask should be set to 255.255.255.0, and the Default Gateway should be set to the VLAN gateway address (e.g., 192.168.10.1 for VLAN 10). The DNS Server should be set to 8.8.8.8, and the Port Status should be enabled. The process should be repeated for all network printers.

5.2.2 Access Points (APs)

Access points were configured to have static IP addresses via the Config tab. The IP address, subnet mask (255.255.255.0), and default gateway for the uplink connection (port 0) were set according to the AP's VLAN ID. The DNS server was set to 8.8.8.8, and the port was enabled. The wireless settings were then configured by specifying the SSID (e.g., HotDesk_WiFi), matching the VLAN ID to the assigned VLAN. This process was repeated for all APs.

5.2.3 IoT Devices

In the 'Config' tab, the correct interface (FastEthernet0/Wireless0) was selected, followed by the selection of 'Static' from the 'IP Configuration' option. An IP address from the 192.168.40.0/24 network range, which is designated for IoT VLAN, was allocated to the device, with the subnet mask, default gateway, and DNS server set to 255.255.255.0, 192.168.40.1, and 8.8.8.8, respectively. The status of the port was checked to ensure it was 'On,' followed by toggling the port OFF/ON to ensure the configuration was successfully uploaded. This was repeated for all the IoT devices to ensure they were reachable from the management VLAN for remote control purposes.

5.3 DHCP Configuration

The DHCP was configured on the main router to automatically assign an IP address to wireless clients in VLAN 20 and VLAN 30. This reduces manual configuration and ensures seamless connectivity for the freelancers and guests.

5.3.1 VLAN 20 Pool (Tenants)

A DHCP pool named TENANTS_POOL was created for the Hot-Desk freelancers on the first floor. The following is the commands used in the main router:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool TENANTS_POOL
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
```

Figure 5.3.1.1: DHCP Configuration for VLAN 20 (Tenants)

5.3.2 VLAN 30 Pool (Guests)

A DHCP pool named GUEST_POOL was created for guest users in the communal lounge. The following is the commands used in the main router:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool GUEST_POOL
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

Figure 5.3.2.1: DHCP Configuration for VLAN 30 (Guests)

5.3.3 Excluded Addresses

Static IP addresses were reserved for the infrastructure devices to prevent conflicts with the DHCP-assigned addresses. The following is the commands used in the main router:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10
Router(config)#end
```

Figure 5.3.3.1: DHCP Configuration for Excluded Addresses

5.4 Router Configuration

5.4.1 Subinterfaces and IP Addressing

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#!
Router(config-if)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)#!
Router(config-subif)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#!
Router(config-subif)#interface gigabitEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#!
Router(config-subif)#interface gigabitEthernet 0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
```

Figure 5.4.1: Subinterfaces and IP Addressing on Main Router

The main router uses a single physical connection to the core switch, but creates four subinterfaces where each is used for a single VLAN. Each subinterface is assigned a VLAN ID using the encapsulation dot1Q command and is given a unique IP address, which is considered the default gateway for devices in that VLAN.

5.4.2 Static Routes

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router(config)#exit
```

Figure 5.4.2.1: Static Routes Configuration on Main Router

On the Main Router, a default static route (ip route 0.0.0.0.0.0.0.0.10.0.0.1) is configured to send all unknown traffic, such as Internet-bound packets to the ISP-Router.

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#!
Router(config-if)#interface gigabitEthernet 0/2
Router(config-if)# ip address 203.0.113.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#ip route 192.168.10.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.40.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
```

Figure 5.4.2.2 :Static Routes Configuration on ISP-Router

On the ISP-Router, static routes are added for each internal VLAN, pointing back to the Main Router via 10.0.0.2. These routes ensure that return traffic from the Internet or external server can reach devices inside the coworking space. Together, these static routes complete the two-way communication path between internal networks and external destinations.

5.5 ACL Configuration

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip access-list extended BLOCK_GUEST
Router(config-ext-nacl)#deny ip any 192.168.10.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.20.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.40.0 0.0.0.255
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#interface gigabitEthernet 0/0.30
Router(config-subif)#interface gigabitEthernet 0/0.30
Router(config-subif)# ip access-group BLOCK_GUEST in
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5.5.1: ACL Configuration BLOCK_GUEST on Main Router

The BLOCK_GUEST ACL is used to secure the guest network (VLAN 30) and prevent direct access to the administration (VLAN 10) and internet of things (VLAN 40) segments without restricting internal communication within the guest VLAN. To maintain basic connectivity tests, it specifically permits ICMP echo-replies from the restricted VLANs to return to guest devices. The "permit ip any any" statement ensures guests retain full access to the internet and external servers.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip access-list extended BLOCK_GUEST
Router(config)#ip access-list extended BLOCK_GUEST
Router(config-ext-nacl)#permit icmp 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255 echo-reply
Router(config-ext-nacl)#deny ip any 192.168.10.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.20.0 0.0.0.255
Router(config-ext-nacl)#deny ip any 192.168.40.0 0.0.0.255
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5.5.2: ACL Configuration BLOCK_TENANT on Main Router

An extended ACL, BLOCK_TENANT, was created to allow DHCP communication and internal communication within the tenant network. ICMP traffic was also permitted to allow connectivity testing to an external host. Statements to deny access from the tenant network to the internal VLANs 10 and 40 were also added. A final permit statement was included to allow other traffic such as Internet access. This configuration ensures that the tenant users are restricted from accessing IoT devices while still being able to obtain IP addresses and access external resources.

5.6 Wireless Configuration

5.6.1 Access Point SSIDs and VLANs

Each Access Point was configured with a unique SSID and mapped to its respective VLAN to verify if there is a proper network segmentation for wireless users

Access Point	Location	SSID	VLAN	Purpose
AP-ADMIN	Ground Floor (Admin Office)	ADMIN_WIFI	10	Secures wireless access for admin laptop and tablet
AP-GUEST	Ground Floor (Communal Lounge)	GUEST_WIFI	30	Provides internet access only up to 15 guest devices
AP-HOT DESK	First Floor (Hot-Desk Area)	HOTDESK_WIFI	20	Supports up to 40 simultaneous freelancer devices
AP-CONFERENCE ROOM	First Floor (Conference Room)	CONFERENCE_WIFI	40	Gives connection to wireless projectors and verifies IoT management for smart lightings

Table 5.6.1: Access Points SSIDs and VLANs

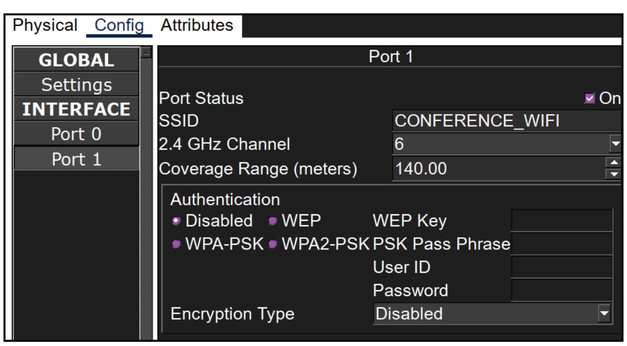

Figure 5.6.1.1: SSID for AP-CONFERENCE ROOM

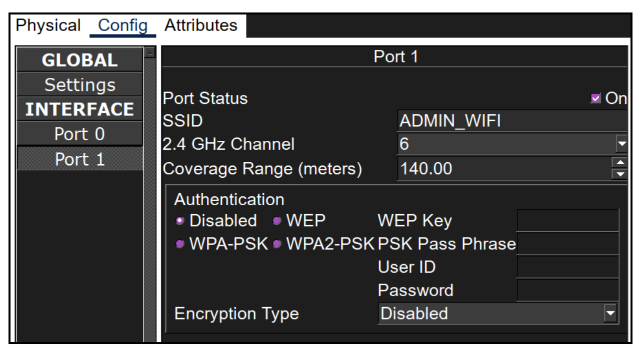

Figure 5.6.1.2: SSID for AP-ADMIN

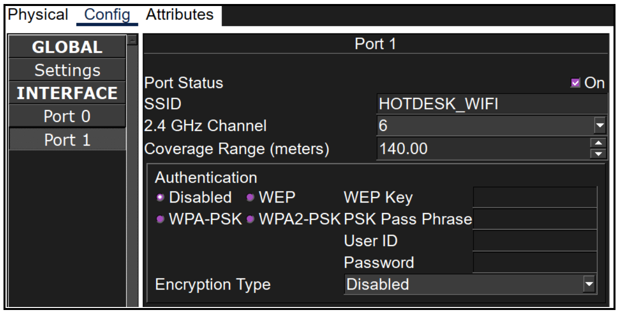

Figure 5.6.1.3: SSID for AP-HOTDESK

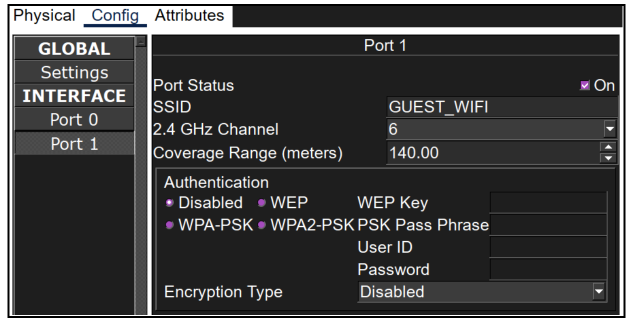

Figure 5.6.1.4: SSID for AP-GUEST

5.7 External Server Configuration

External server was configured to simulate internet services and its to verify an end to end connectivity from internal networks

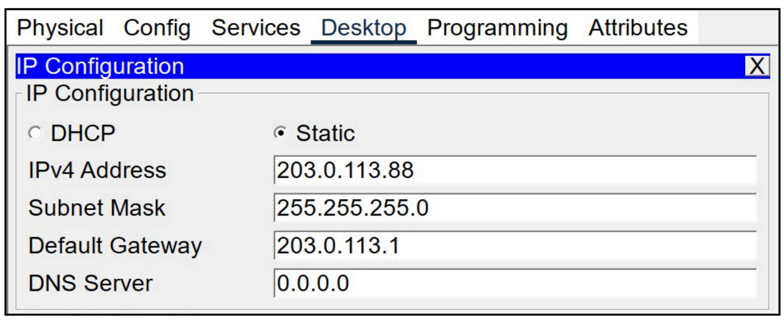

Figure 5.7.1: Services on External Server

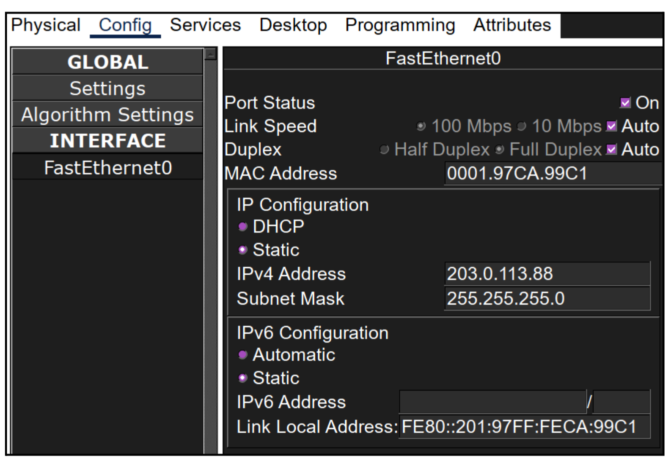

Figure 5.7.2: Config tab on External Server

6.0 Troubleshooting Details

This section of the report verifies the functionality of the network using the *ping* command.

6.1 Admin PC Connectivity

The Admin PC, which belongs to VLAN 10 is configured with full network access privileges. This then allows the Admin PC to communicate with the external server, IoT devices, tenant PCs and guest PCs.

```
C:\>ping 203.0.113.88

Pinging 203.0.113.88 with 32 bytes of data:

Reply from 203.0.113.88: bytes=32 time<1ms TTL=126
Reply from 203.0.113.88: bytes=32 time<1ms TTL=126
Reply from 203.0.113.88: bytes=32 time<1ms TTL=126
Reply from 203.0.113.88: bytes=32 time<1ms TTL=126

Ping statistics for 203.0.113.88:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 6.1.1: Admin Pinging to External Server

```
C:\>ping 192.168.40.16

Pinging 192.168.40.16 with 32 bytes of data:

Reply from 192.168.40.16: bytes=32 time<1ms TTL=254
Reply from 192.168.40.16: bytes=32 time<1ms TTL=254
Reply from 192.168.40.16: bytes=32 time<1ms TTL=254
Reply from 192.168.40.16: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.40.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 6.1.2: Admin Pinging to CCTV-FF-1

```
C:\>ping 192.168.40.20

Pinging 192.168.40.20 with 32 bytes of data:

Reply from 192.168.40.20: bytes=32 time<1ms TTL=254
Reply from 192.168.40.20: bytes=32 time<1ms TTL=254
Reply from 192.168.40.20: bytes=32 time=10ms TTL=254
Reply from 192.168.40.20: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.40.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Figure 6.1.3 Admin Pinging to Admin Room RFID

```
C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time=17ms TTL=127

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

Figure 6.1.4 Admin Pinging to Tenant PC-4

```
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time=6ms TTL=127
Reply from 192.168.30.11: bytes=32 time=4ms TTL=127
Reply from 192.168.30.11: bytes=32 time=5ms TTL=127
Reply from 192.168.30.11: bytes=32 time=6ms TTL=127

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 6ms, Average = 5ms
```

Figure 6.1.5: Admin Pinging to Guest

```
C:\>ping 192.168.10.13

Pinging 192.168.10.13 with 32 bytes of data:

Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 6.1.6: Admin Pinging to Receptionist

6.2 Receptionists PC Connectivity

The receptionists PCs which consist of three computers are assigned to VLAN 10, it is configured with the same network access privileges as the Admin PC. This allows the receptionists to access all relevant network resources, including the external server, IoT devices and printer located on the Ground Floor.

```
C:\>ping 203.0.113.88

Pinging 203.0.113.88 with 32 bytes of data:

Reply from 203.0.113.88: bytes=32 time<1ms TTL=126
Reply from 203.0.113.88: bytes=32 time<1ms TTL=126
Reply from 203.0.113.88: bytes=32 time<1ms TTL=126
Reply from 203.0.113.88: bytes=32 time<1ms TTL=126

Ping statistics for 203.0.113.88:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 6.2.1: Receptionist Pinging to External Server

```
C:\>ping 192.168.40.20

Pinging 192.168.40.20 with 32 bytes of data:

Reply from 192.168.40.20: bytes=32 time<1ms TTL=254
Reply from 192.168.40.20: bytes=32 time<1ms TTL=254
Reply from 192.168.40.20: bytes=32 time=10ms TTL=254
Reply from 192.168.40.20: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.40.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Figure 6.2.2: Receptionist Pinging to CCTV-FF-1

```
C:\>ping 192.168.10.50

Pinging 192.168.10.50 with 32 bytes of data:

Reply from 192.168.10.50: bytes=32 time<1ms TTL=128
Reply from 192.168.10.50: bytes=32 time<1ms TTL=128
Reply from 192.168.10.50: bytes=32 time=1ms TTL=128
Reply from 192.168.10.50: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 6.2.3: Receptionist Pinging to Reception Printer

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=12ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=3ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 4ms
```

Figure 6.2.4: Receptionist Pinging to Tenant PC

```
C:\>ping 192.168.10.52

Pinging 192.168.10.52 with 32 bytes of data:

Reply from 192.168.10.52: bytes=32 time<1ms TTL=128
Reply from 192.168.10.52: bytes=32 time<1ms TTL=128
Reply from 192.168.10.52: bytes=32 time<1ms TTL=128
Reply from 192.168.10.52: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.52:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 6.2.4: Receptionist Pinging to Admin PC

6.3 Guest Device Connectivity

The Guest devices, which are connected wirelessly and assigned to VLAN 30, are configured with restricted network access. Guests are only permitted to access the external server. Other access to internal networks such as IoT devices, Receptionist PCs, Tenant PCs, and the Admin PCs is denied to ensure isolation.

```
C:\>ping 203.0.113.88

Pinging 203.0.113.88 with 32 bytes of data:

Reply from 203.0.113.88: bytes=32 time=470ms TTL=126
Request timed out.
Reply from 203.0.113.88: bytes=32 time=13ms TTL=126
Request timed out.

Ping statistics for 203.0.113.88:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 470ms, Average = 241ms
```

Figure 6.3.1: Guest Pinging to External Server

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 6.3.2: Guest Pinging to Admin PC

```
C:\>ping 192.168.20.50

Pinging 192.168.20.50 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.20.50:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 6.3.3: Guest Pinging to Tenant PC

```
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 6.3.4: Guest Pinging to IP Camera (CCTV-FF-1)

```
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time=69ms TTL=128
Reply from 192.168.30.11: bytes=32 time=53ms TTL=128
Reply from 192.168.30.11: bytes=32 time=15ms TTL=128
Reply from 192.168.30.11: bytes=32 time=23ms TTL=128

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 15ms, Maximum = 69ms, Average = 40ms
```

Figure 6.3.5: Guest Pinging to another Guest Device

6.4 Tenant PC Connectivity

Premium Tenant devices which belong to VLAN 20 are granted access to external server and is able to communicate with other tenant devices as well as the printer on the First Floor. However, they are denied access to Guest PCs, IoT devices and Receptionists PCs.

```
C:\>ping 203.0.113.88

Pinging 203.0.113.88 with 32 bytes of data:

Reply from 203.0.113.88: bytes=32 time=470ms TTL=126
Request timed out.
Reply from 203.0.113.88: bytes=32 time=13ms TTL=126
Request timed out.

Ping statistics for 203.0.113.88:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 470ms, Average = 241ms
```

Figure 6.4.1: Tenant PC Pinging to External Server

```
C:\>ping 192.168.20.50

Pinging 192.168.20.50 with 32 bytes of data:

Reply from 192.168.20.50: bytes=32 time=1ms TTL=128
Reply from 192.168.20.50: bytes=32 time<1ms TTL=128
Reply from 192.168.20.50: bytes=32 time<1ms TTL=128
Reply from 192.168.20.50: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 6.4.2: Tenant PC Pinging to Printer Private Office

```
C:\>ping 192.168.20.7

Pinging 192.168.20.7 with 32 bytes of data:

Reply from 192.168.20.7: bytes=32 time=2ms TTL=128
Reply from 192.168.20.7: bytes=32 time<1ms TTL=128
Reply from 192.168.20.7: bytes=32 time<1ms TTL=128
Reply from 192.168.20.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Figure 6.4.3: Tenant PC Pinging to Tenant PC 6

```
C:\>ping 192.168.20.51

Pinging 192.168.20.51 with 32 bytes of data:

Reply from 192.168.20.51: bytes=32 time<1ms TTL=128
Reply from 192.168.20.51: bytes=32 time=3ms TTL=128
Reply from 192.168.20.51: bytes=32 time<1ms TTL=128
Reply from 192.168.20.51: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.51:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

Figure 6.4.4: Tenant PC Pinging to Hot Desk Printer

```
C:\>ping 192.168.30.12

Pinging 192.168.30.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 6.4.5: Tenant PC Pinging to Guest Laptop

```
C:\>ping 192.168.10.13

Pinging 192.168.10.13 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.13:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 6.4.6: Tenant PC Pinging to Receptionist PC2

```
C:\>ping 192.168.40.11

Pinging 192.168.40.11 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.40.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 6.4.7: Tenant PC Pinging CCTV-FF-2

7.0 Conclusion

In conclusion, the smart co-working space successfully provides secure connectivity for staff, tenants, guests and IoT devices. By using Cisco Packet Tracer, the network was designed using all the required hardware which is routers, switches, wireless access points and as well as end devices that supports both wired and wireless communication for Ground Floor and First Floor.

VLANs were implemented to separate different user groups which are Administration, Tenants, Guests, and IoT devices. DHCP services are implemented to assign IP addresses to wireless users and tenants devices while static IP addressing was used for printers, access points and IoT devices. This is done to ensure consistency in network management.

Furthermore, the network also integrated smart facilities such as RFID door locks, IP surveillance cameras, smart lighting and wireless projectors. To verify the connection, ping tests were done in CLI to confirm that the administration and receptionists could manage IoT devices and access printers, tenants could access their resources and the Internet, and guest users were restricted to only Internet access.

Overall, the designed network successfully fulfills the project requirements as it delivers a feasible, well-segmented infrastructure that supports both traditional networking devices and modern IoT facilities in the smart co-working space.

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